IIOT NO 5 ROLL NO 31 B

```
import pandas as pd
import matplotlib.pyplot as plt
data={
"Machine":["M1","M2","M3","M4","M5","M6","M7","M8","M9","M10"],
"Temperature":[25,28,31,45,27,29,42,26,30,47],
"Pressure":[100,102,98,110,101,99,108,97,103,112],
"Vibration":[2.1,2.5,2.8,4.5,2.3,2.6,4.2,2.0,2.7,4.8]
}
df=pd.DataFrame(data)
print("INDUSTRIAL SENSOR DATA")
print(df)
print("\nSTATISTICAL ANALYSIS")
print("Average Temperature:",df["Temperature"].mean())
print("Maximum Temperature:",df["Temperature"].max())
print("Minimum Temperature:",df["Temperature"].min())
print("Average Pressure:",df["Pressure"].mean())
print("Average Vibration:",df["Vibration"].mean())
threshold=40
abnormal=df[df["Temperature"]>threshold]
print("\nABNORMAL TEMPERATURE READINGS")
print(abnormal)
# Temperature graph
plt.figure(figsize=(8,4))
plt.plot(df["Machine"],df["Temperature"],marker="o")
plt.axhline(y=threshold,linestyle="--",label="Temperature Limit")
plt.title("Industrial Machine Temperature Analysis")
plt.xlabel("Machine")
plt.ylabel("Temperature (°C)")
plt.legend()
plt.grid()
plt.tight_layout()
plt.show()
# Vibration graph
plt.figure(figsize=(8,4))
plt.bar(df["Machine"],df["Vibration"])
```

```
plt.title("Industrial Machine Vibration Analysis")
plt.xlabel("Machine")
plt.ylabel("Vibration Level")
plt.grid(axis="y")
plt.tight_layout()
plt.show()
```

OUTPUT :

INDUSTRIAL SENSOR DATA

	Machine	Temperature	Pressure	Vibration
0	M1	25	100	2.1
1	M2	28	102	2.5
2	M3	31	98	2.8
3	M4	45	110	4.5
4	M5	27	101	2.3
5	M6	29	99	2.6
6	M7	42	108	4.2
7	M8	26	97	2.0
8	M9	30	103	2.7
9	M10	47	112	4.8

STATISTICAL ANALYSIS

Average Temperature: 33.0

Maximum Temperature: 47

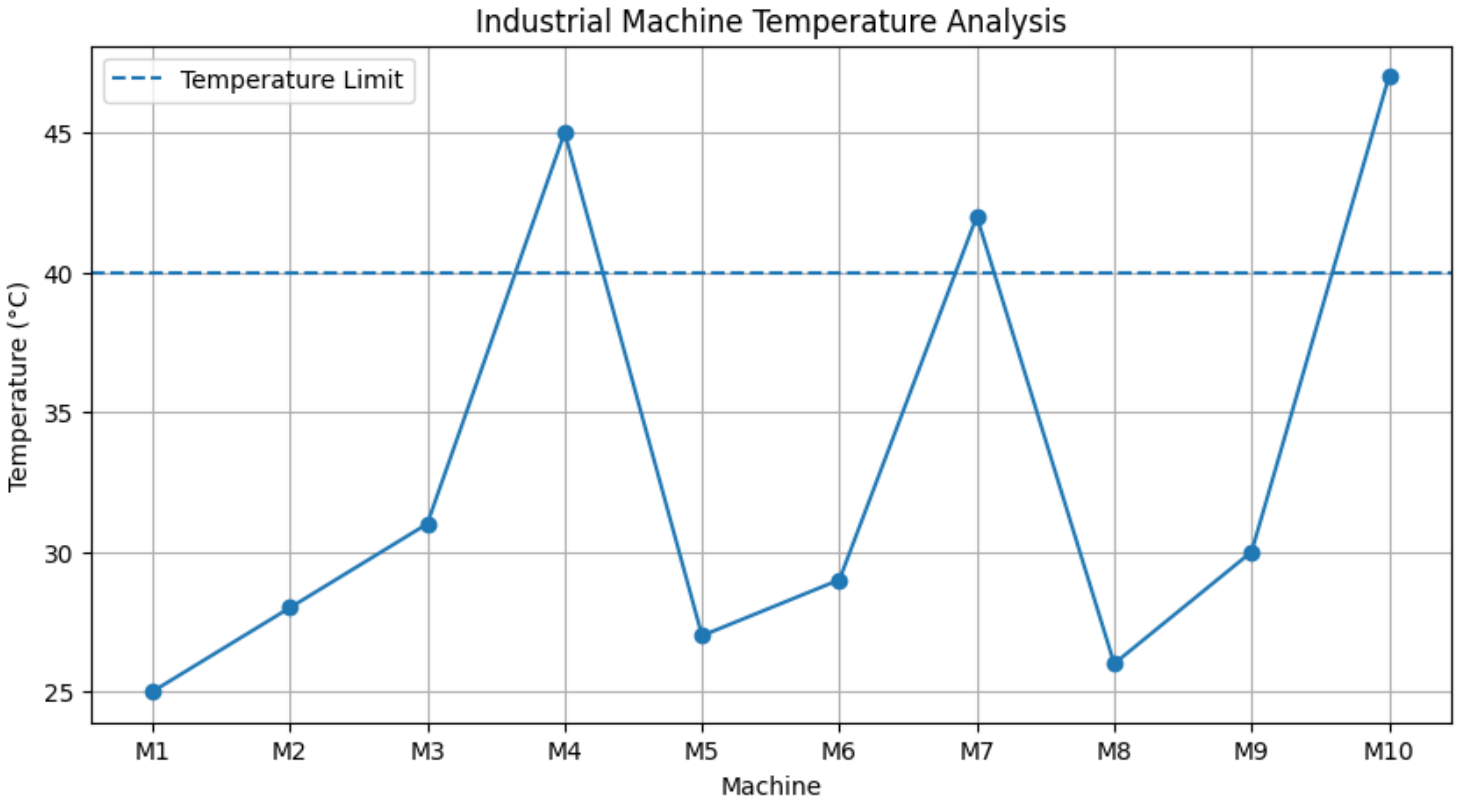
Minimum Temperature: 25

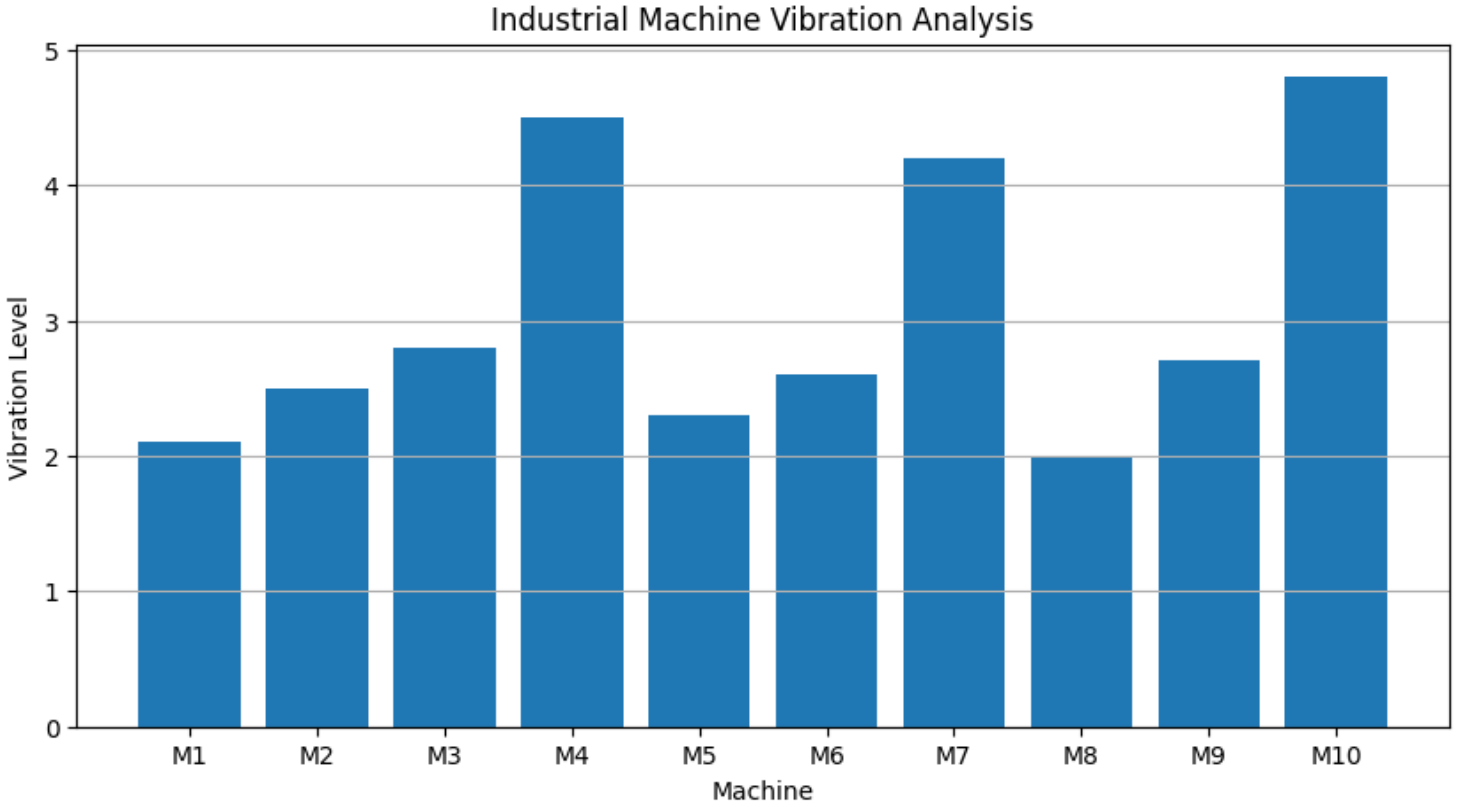
Average Pressure: 103.0

Average Vibration: 3.05

ABNORMAL TEMPERATURE READINGS

	Machine	Temperature	Pressure	Vibration
3	M4	45	110	4.5
6	M7	42	108	4.2
9	M10	47	112	4.8


Industrial Machine Temperature Analysis
Temperature Limit
Temperature (°C)
45
40
35
30
25
M1
M2
M3
M4
M5
M6
M7
M8
M9
M10
Machine


Industrial Machine Vibration Analysis
Vibration Level
5
4
3
2
1
0
M1
M2
M3
M4
M5
M6
M7
M8
M9
M10
Machine